Informativity and Identifiability for Identification of Networks of Dynamical Systems

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***Abstract*— We study informativity and network identifiability for dynamic networks with unknown input and measurement transfer functions, constrained entries, and repeated open-loop transfer functions. Positive definiteness of the external-signal spectrum and full generic row rank of the transfer function matrix from external signals to the predictor inputs give a sufficient condition for generic informativity. We determine this rank by disjoint paths for independent entries and by a Gröbner basis test for constrained entries. The dimension of the generic fiber of the map from open-loop to closed-loop transfer functions determines whether finitely many networks are compatible with the same closed-loop transfer function. A repeated-entry example has two admissible networks in a generic fiber, an ambiguity that generic Jacobian rank alone does not count.**

***Index Terms*— System Identification, Networks, Dynamical Systems, Informativity, Generic local network identifiability.**

I. Introduction

SYSTEM identification for networks of dynamical systems is an active research field with a long history. Some early references are [1]–[3]. Some of the work has been focusing on identifying the full network, e.g. [3]–[5], while other work has been focusing on identifying parts of the network, or *subnetworks*, e.g. [2], [6]–[11]. The latter is an example of system identification under feedback, a topic that also has a long history, see e.g. [12].

Prediction error methods use a predictor to estimate an output from measured signals. Their analysis raises two distinct questions. Informativity asks whether the data determine the predictor within a candidate model class. Network identifiability asks whether the resulting closed-loop transfer functions determine the open-loop transfer functions in the network. The latter question differs from identifiability of a particular numerical parametrization. We study generic properties of the specified model class: properties that can fail only on a proper algebraic subset of its admissible networks. In particular, generic local network identifiability in the algebraic sense allows finitely many networks to share the same closed-loop transfer function. This is related to structural identifiability, which asks whether the parameters of a specified dynamical model can be recovered from its input-output behavior; Ljung and Glad study global identifiability for general nonlinear model structures using differential algebra [13]. Our identifiability target is instead the unknown open-loop transfer functions of a specified network structure, and our generic local criterion permits finite ambiguity.

Prior work addresses partial node measurement [3], [14], [15] and data informativity [2], [16]. The novel aspects of this paper relative to these and the cited rank and identifiability results are:

- The novel model scope is the joint treatment of unknown input and measurement transfer functions, direct input-to-output paths, and repeated or constrained transfer functions. The network models in [3], [14] use fixed input and measurement selections.
- The novel aspect of the rank analysis is its application to this model: disjoint paths include direct input-to-output paths, while polynomial elimination handles known relations among entries. The path-rank principle [17] and spectral informativity implication [18] are established results.
- The novel identifiability contribution is applying the standard algebraic dimension criterion [19] to the map from open-loop to closed-loop transfer functions for this network class, including repeated and constrained entries, and solving the resulting equations to expose finite ambiguity. Example 3 demonstrates this with two compatible networks, which generic Jacobian rank alone does not count.

II. Model of Network

We consider a linear dynamic network model defined by

$$\begin{aligned} P(q)w_k &= Q(q)u_k \\ z_k &= R(q)w_k + S(q)u_k \end{aligned} \tag{1}$$

where (P, Q, R, S) are rational transfer function matrices in the forward shift operator q. The dimensions of these transfer function matrices are $n \times n$, $n \times m$, $p \times n$, and $p \times m$, respectively. This also defines the dimensions of all the signals, where u_k is referred to as the input signal, z_k as the output signal, and w_k as the node signal. The familiar model $w_k = G(q)w_k + B(q)u_k$, $z_k = C(q)w_k$, where G contains node-to-node transfer functions, B input-to-node transfer functions, and C node-to-output transfer functions, is obtained with $P = I - G$, $Q = B$, $R = C$, and $S = 0$. Thus P describes internal connections, Q describes how inputs

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enter the nodes, R describes how nodes are measured, and S describes paths from inputs to outputs that bypass the nodes. The model allows the input-to-node transfer functions in Q and the node-to-output transfer functions in R to be unknown. It also allows direct input-to-output paths through S and known relations among entries of P, Q, R, S. The form $Pw = Qu$ retains the coefficients of the original node equations when normalization would introduce shared factors into the input and connection transfer functions. Filtered or combined sensor readings give R, while direct input effects on measured outputs give S. For example, physical networks with local node dynamics and diffusive couplings can be written $D(q)w_k = K(q)w_k + B(q)u_k$, where diagonal D contains the local dynamics and K the couplings [20]. Here $P = D - K$ retains these physical coefficients and $Q = B$ retains the separate input matrix. If D is invertible, normalization gives $G = D^{-1}K$ and an input transfer function matrix $D^{-1}B$, but these entries then share unknown factors and relations inherited from D. The constrained-entry tests below retain these relations. The notation also accommodates a discrete-time descriptor model $Ew_{k+1} = Aw_k + Bu_k$, $z_k = Cw_k + Du_k$, where E, A, B, C, D are real matrices and $\det(qE - A)$ is not identically zero. It has $P(q) = qE - A$, $Q = B$, $R = C$, and $S = D$. When E is singular, this representation retains the algebraic equations alongside the state dynamics. Prescribed zero patterns and known coefficients in these matrices specify a constrained model class. Its unknown matrix entries are real constants, and the coefficients of $P(q) = qE - A$ retain this structure. We assume that the structural zero pattern and any known entries or relations among unknown entries of (P, Q, R, S) are specified. The task is to determine these entries for the given model class and zero pattern. Structurally free entries may take the value zero at exceptional realizations. We assume that P is generically invertible: $\det P$, viewed as a rational expression in its free entries, is not identically zero. A generic choice of these entries then gives an invertible rational matrix P. We call a realization *well-posed* when its P is invertible and apply formulas involving P^{-1} only to such realizations. When evaluating a transfer function, we use $\lambda \in \mathbb{C}$ as its complex argument, for example $P(\lambda)$. For a well-posed realization, $P(\lambda)$ is invertible at all but finitely many λ where its entries are defined. Stability is assumed separately when needed. Generic invertibility is checked from P in each example. Let θ denote the unknown parameters and θ_0 their values in the true network. When needed, we write $P(q;\theta)$ and $y_k(\theta)$ to show parameter dependence. We otherwise omit the arguments q and θ.

III. Innovation form and Predictor

We partition u_k into a measured input r_k and an unmeasured noise input e_k. We write $u_k = (r_k, e_k)$, where r_k has m_r components and e_k has m_e components. We also partition $Q(q)$ and $S(q)$ conformably as

$$Q(q) = \begin{bmatrix} Q_r(q) & Q_e(q) \end{bmatrix}, \quad S(q) = \begin{bmatrix} S_r(q) & S_e(q) \end{bmatrix}.$$

We assume that the signals available to the predictor form $z_k = (y_k, r_k)$. Here y_k has p_y components, and r_k has m_r components, so $p = p_y + m_r$. The measured outputs y_k and inputs r_k specify which predictor is considered. For a subnetwork, measured node signals at its boundary can also serve as measured inputs to that subnetwork. We conformally write

$$\begin{bmatrix} R(q) & S(q) \end{bmatrix} = \begin{bmatrix} R(q) & S_r(q) & S_e(q) \end{bmatrix} = \begin{bmatrix} R_y(q) & S_{yr}(q) & S_{ye}(q) \\ 0 & I & 0 \end{bmatrix}$$

for appropriately defined transfer function matrices. The resulting closed-loop relation from u_k to y_k is

$$y_k = G_c(q)r_k + H_c(q)e_k \tag{2}$$

where

$$\begin{aligned} \begin{bmatrix} G_c(q) & H_c(q) \end{bmatrix} &= R_y(q)P(q)^{-1} \begin{bmatrix} Q_r(q) & Q_e(q) \end{bmatrix} \\ &\quad + \begin{bmatrix} S_{yr}(q) & S_{ye}(q) \end{bmatrix}. \end{aligned} \tag{3}$$

Assume that the measured output has an innovation representation

$$\hat{y}_k = G_c(q)r_k + G_o(q)\epsilon_k \tag{4}$$

$$y_k = \hat{y}_k + \epsilon_k \tag{5}$$

where $G_o(q)$ is a transfer function matrix and the innovations ϵ_k are independent, identically distributed zero-mean random vectors. Such a representation can require solving an algebraic Riccati equation [21]. Only its existence is used here.

From the above equations we realize that we may write

$$\hat{y}_k = W(q)z_k \tag{6}$$

where

$$W(q) = \begin{bmatrix} W_y(q) & W_r(q) \end{bmatrix} = (I + G_o(q))^{-1} \begin{bmatrix} G_o(q) & G_c(q) \end{bmatrix}.$$

We assume that this predictor for y_k is stable, meaning that all poles of $W(q)$ lie strictly inside the unit circle; this is checked in each example. Solving the displayed relation for G_c gives

$$G_c(q) = (I - W_y(q))^{-1} W_r(q). \tag{7}$$

Thus the predictor for the selected signals determines the corresponding closed-loop transfer function G_c. The selection therefore specifies both the predictor and the transfer function to identify. When $R_y = I$, all node signals are measured. When its rows select individual node signals, only those nodes are measured. The model also permits a measurement equation with unknown measurement transfer functions, as in Example 3. The input matrices Q_r and Q_e may likewise contain unknown input transfer functions rather than fixed input selections.

IV. Informativity

Throughout, "almost all frequencies" means outside a set of Lebesgue measure zero on $[-\pi, \pi]$. We investigate when the predictor can be determined from the measured signals. Predictor uniqueness is the data requirement for prediction-error estimation; recovering open-loop transfer functions from that predictor is a separate question. We adapt the prediction-error definition of data informativity in [16]:

Definition 1: The signal z_k is said to be *informative enough* if for any two parameters θ_1 and θ_2 such that the closed loop system is stable, it holds that

$$\mathbb{E}\left[\left(\hat{y}_k(\theta_1) - \hat{y}_k(\theta_2)\right) \left(\hat{y}_k(\theta_1) - \hat{y}_k(\theta_2)\right)^T \right] = 0 \tag{8}$$

implies that $W\left(e^{i\omega};\theta_1\right) = W\left(e^{i\omega};\theta_2\right)$ for almost all ω. It holds that

$$\hat{y}_k(\theta_1) - \hat{y}_k(\theta_2) = \Delta W(q) z_k$$

where $\Delta W(q) = W(q;\theta_1) - W(q;\theta_2)$. From Parseval's formula, it follows that (8) is equivalent to

$$\Delta W\left(e^{i\omega}\right) \Phi_z(\omega) \Delta W\left(e^{-i\omega}\right)^T = 0$$

for almost all ω, where Φ_z is the spectrum of z_k. Hence, a sufficient condition for the above equality to imply that $W\left(e^{i\omega};\theta_1\right) = W\left(e^{i\omega};\theta_2\right)$ for almost all ω, is that $\Phi_z(\omega)$ is positive definite for almost all ω.

The closed-loop transfer function from $u_k = (r_k, e_k)$ to z_k is

$$\Pi(q) = R(q)P(q)^{-1}Q(q) + S(q)$$

If the spectrum of u_k is positive definite and $\Pi\left(e^{i\omega};\theta_0\right)$ has full row rank for almost all ω, then the spectrum of z_k is positive definite. Positive definiteness of the predictor-input spectrum is a standard sufficient condition for informativity [18]; related network-specific excitation conditions are studied in [16], [22]. Here Π describes how the external signals reach the selected predictor inputs. The full-row-rank condition depends on the true network at θ_0. To obtain a condition for generic true networks, let

$$M(q) = \begin{bmatrix} P(q) & Q(q) \\ -R(q) & S(q) \end{bmatrix}. \tag{9}$$

As in [16], we investigate the *generic* rank

$$\operatorname{rank}_g \Pi = \max_{M \text{ in the model class}} \operatorname{rank} \Pi.$$

The rank is taken over the field of rational transfer functions. For the free-entry results below, the free entries of M are algebraically independent, meaning that they satisfy no nonzero polynomial relation, over $\mathbb{C}(\lambda)$, the field of rational functions in λ. Constraints are treated separately. For algebraically generic admissible values of the free entries, the rank of Π over $\mathbb{C}(\lambda)$ equals $\operatorname{rank}_g \Pi$. Since its minors are rational functions of λ, this rank also holds at all but finitely many λ where Π is defined. This gives a sufficient condition for informativity at algebraically generic true networks.

Theorem 1: Under the standing assumptions on the network and the candidate predictors, suppose that the spectrum of $u_k = (r_k, e_k)$ is positive definite for almost all frequencies and that Π has full generic row rank. Then, for algebraically generic admissible choices of the true network's free transfer functions, the resulting signal z_k is informative enough.

Proof: For algebraically generic admissible choices of the true network's free transfer functions, full generic row rank implies that $\Pi(e^{i\omega};\theta_0)$ has full row rank for almost all ω. Since

$$\Phi_z(\omega) = \Pi(e^{i\omega};\theta_0)\Phi_u(\omega)\Pi(e^{i\omega};\theta_0)^*,$$

where * denotes the conjugate transpose, the spectrum of z_k is then positive definite for almost all frequencies. Informativity follows from the preceding argument. ∎

Exact cancellations may reduce the rank of Π at exceptional true networks. For each true network covered by the theorem, the definition of informativity still applies to every pair of admissible candidate predictors. The generic rank is understood relative to the model class under consideration, respecting any known transfer functions or imposed constraints.

Remark 1: If some components of u_k are set to zero, the corresponding columns of $M(q)$ may be removed. If some predictor columns are fixed across all candidate models, the measurement rows of $M(q)$ corresponding to their predictor inputs may be omitted when testing informativity for the remaining columns. Merely being uninterested in unknown predictor columns does not justify this removal, since changes in those columns may cancel changes in the columns of interest. More generally, known relationships between predictor columns may allow the predictor to be expressed using fewer, known combinations of the measured signals. Informativity can then be investigated for this reduced predictor-input vector. We next give graph and Gröbner basis conditions for the generic rank of Π.

A. Graphical Condition

Since we are investigating generic properties, we drop the dependence on the forward shift operator q or the complex argument λ. For the graph construction, we assume that the diagonal entries of P are structurally nonzero, as for $sI - A$ in [17]. Since P is generically invertible, this can always be achieved by a row permutation. Applying the same permutation to Q leaves $RP^{-1}Q + S$ unchanged. In this subsection, P, Q, and M refer to the matrices after any such permutation.

Block elimination gives

$$\begin{bmatrix} I & 0 \\ RP^{-1} & I \end{bmatrix} M \begin{bmatrix} I & -P^{-1}Q \\ 0 & I \end{bmatrix} = \begin{bmatrix} P & 0 \\ 0 & RP^{-1}Q + S \end{bmatrix}.$$

For each well-posed network, P is invertible. Hence, the rank of M is the same as the rank of Π plus the dimension of P, which is n. Van der Woude [17] characterizes the generic rank of $C(sI - A)^{-1}B$ by vertex-disjoint paths. We give the corresponding argument for $\Pi = RP^{-1}Q + S$ below. For independent free entries of M, rank drops below its generic value only on a proper affine variety, a polynomial zero set smaller than the full parameter space, [23].

Define a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, where $\mathcal{V} = \mathcal{U} \cup \mathcal{W} \cup \mathcal{Z}$ with $\mathcal{U} = \{\mathcal{U}_1, \ldots, \mathcal{U}_m\}$, $\mathcal{W} = \{\mathcal{W}_1, \ldots, \mathcal{W}_n\}$, $\mathcal{Z} = \{\mathcal{Z}_1, \ldots, \mathcal{Z}_p\}$. The first n columns of M correspond to $\mathcal{W}_1, \ldots, \mathcal{W}_n$ and the next m to $\mathcal{U}_1, \ldots, \mathcal{U}_m$; its rows correspond to $\mathcal{W}_1, \ldots, \mathcal{W}_n, \mathcal{Z}_1, \ldots, \mathcal{Z}_p$. For every structurally nonzero entry M_{v_j, v_i}, include the directed edge (v_i, v_j), except when $v_i = v_j$. A path is a sequence of vertices joined by directed edges. Paths from $\mathcal{U}$ to $\mathcal{Z}$ are *disjoint* if they share no vertices.

We are now able to state the following result, which is a generalization of Theorem 2 in [17].

Theorem 2: Assume that all structurally nonzero entries of M are independent free variables. Then the maximum number of disjoint paths in $\mathcal{G}$ from $\mathcal{U}$ to $\mathcal{Z}$ is equal to the generic rank of Π.

Proof: Choose k input columns J and output rows I. The Schur-complement identity gives

$$\det \begin{bmatrix} P & Q_J \\ -R_I & S_{I,J} \end{bmatrix} = \det(P) \det(\Pi_{I,J}).$$

In a determinant term, each selected input has one outgoing edge and each selected output has one incoming edge. Each state vertex has at most one edge of either kind, so the term contains k disjoint input-to-output paths; its other edges form cycles or diagonal factors. An entry of S gives a one-edge path. Conversely, k disjoint paths select a determinant term by using diagonal entries of P at unused state vertices. Distinct selections give distinct monomials because the structurally nonzero entries are independent; hence this term cannot cancel identically. The equivalence holds for every k. ■

Remark 2: The problem of finding the maximum number of disjoint paths can be formulated as a maximum flow problem for an associated flow graph.

B. Gröbner Basis Condition

We next investigate the generic rank of Π using a Gröbner basis. We still regard Π as a rational function of the free entries of M. Let D_i be the least common multiple of the denominators of the ith row of Π. Multiply each row of Π with this polynomial to obtain the polynomial matrix $\tilde{\Pi}$. Then the generic rank of Π, when well-posedness holds, is the same as the generic rank of $\tilde{\Pi}$ when $D_i \neq 0$. The ideal generated by a collection of polynomials consists of all their polynomial combinations. Define the ideal $\mathcal{I}_k$ generated by all the minors of $\tilde{\Pi}$ of dimension k together with the polynomial $1 - tD$, where $D = \det(P) \prod_i D_i$ and t is an additional variable. Any additional denominator polynomials required by rational constraints or parameterizations are also included as factors in D. The equation $1 - tD = 0$ has a solution in t precisely when $D \neq 0$. All rank statements and solution sets below are understood on this admissible domain. We use a lexicographic order with t as the largest variable. Eliminating t means retaining basis polynomials that do not involve t; their zeros, restricted to $D \neq 0$, give the rank-deficient set. Thus t excludes undefined network values before the rank condition is projected onto the free entries.

Theorem 3: For the ideal $\mathcal{I}_k$ defined above, one of the following holds:

1) If the reduced Gröbner basis is $\{1\}$, then $\operatorname{rank} \Pi \geq k$ at every admissible point.
2) If the basis gives no nonzero polynomial independent of t, then $\operatorname{rank} \Pi < k$ at every admissible point.
3) Otherwise $\operatorname{rank} \Pi \geq k$ generically. The basis polynomials independent of t, together with $D \neq 0$, describe exactly the admissible points where $\operatorname{rank} \Pi < k$. For $k = \operatorname{rank}_g \Pi$, these are the points where rank falls below its generic value.

Proof: On $D \neq 0$, multiplying the rows by D_i does not change rank. Thus $\operatorname{rank} \Pi < k$ exactly when all the k-minors of $\tilde{\Pi}$ vanish. The equation $1 - tD = 0$ keeps precisely the points with $D \neq 0$. Eliminating t gives the closure of their rank-deficient set; imposing $D \neq 0$ again gives the set itself. The three cases follow according to whether this set is empty, fills the admissible component, or is a proper subset of it. ■

Remark 3: Without the restriction $D \neq 0$, the basis polynomials independent of t may also vanish at inadmissible points where $D = 0$.

Remark 4: Known entries of M enter as coefficients in $\mathbb{C}(\lambda)$.

Remark 5: In case some of the entries of M are constrained by rational equations, we first clear their denominators and add the resulting polynomial equations to the ideal, including those denominators as factors in D. Apply the theorem separately to each irreducible component that meets $D \neq 0$. In item 2, "no nonzero polynomial independent of t" means no additional condition beyond the equations already defining that component. Repeated entries can instead be represented by a single variable.

The Gröbner basis condition accommodates known and constrained entries. The graphical condition assumes independent free entries.

V. Identifiability

Under informativity, the predictor determines the closed-loop transfer function G_c through (7). We now ask whether G_c determines the unknown entries of (P, Q_r, R_y, S_{yr}). By (3),

$$G_c = R_y P^{-1} Q_r + S_{yr}$$

Let X be the k-dimensional vector of free entries of these matrices. To count the networks compatible with a given G_c, we make the dependence on X explicit and define the closed-loop map

$$F(X) = R_y(X) P(X)^{-1} Q_r(X) + S_{yr}(X).$$

We keep λ symbolic and regard F as a rational map over $K = \mathbb{C}(\lambda)$. Known rational transfer functions enter as coefficients. For the algebraic analysis, we work over an algebraic closure of K and call sets defined by polynomial equations *varieties*. Their dimension is the maximum number of algebraically independent coordinates. A variety is *irreducible* if it cannot be written as the union of two smaller varieties. Let V_o be the subset of k-tuples on which P is invertible and all expressions defining F are well-defined. Rational realizations satisfying these algebraic conditions correspond to points of V_o whose coordinates belong to K. For these points, equality under F means equality of rational functions, not equality at a fixed complex argument. Let $V_c = F(V_o)$ be the image of F. It need not itself be a variety. We use $\dim(V_c)$ to denote the dimension of its Zariski closure, the smallest variety containing V_c. Here and below, a property holds generically if it holds outside a proper algebraic subset of the relevant irreducible variety. We assume that V_o is irreducible. This holds when the k free entries have no relations beyond the exclusions used to define V_o. The closure may contain points with no admissible preimage; the fiber criterion concerns generic points of the actual image V_c.

Legat and Hendrickx [14] ask whether generic local network identifiability implies generic network identifiability for the model $w = Gw + Br$, $y = Cw$, where B and C select

nodes and the unknowns are independent entries of G. Here, unknown input and measurement transfer functions may also enter Q_r and R_y, including as repeated entries.

Definition 2: We say that we have generic local network identifiability in the algebraic sense if, for generic $G_c \in V_c$, the fiber defined by

$$F(X) = G_c, \qquad X \in V_o, \tag{10}$$

has dimension zero. We use this algebraic meaning of generic local network identifiability throughout the following sections.

Remark 6: A zero-dimensional fiber consists of finitely many isolated algebraic solutions. Thus, generic local network identifiability allows a finite ambiguity, whereas generic network identifiability requires uniqueness. It gives no uniqueness guarantee for a specified realization, including one on an exceptional algebraic subset. In particular, there are only finitely many admissible solutions whose open-loop transfer functions are rational. The algebraic solutions need not all have rational coordinates, and any additional requirements such as properness or stability must also be checked when interpreting an alternative solution as an admissible network. Legat and Hendrickx [14] formulate local identifiability using complex values of transfer functions at a single complex argument λ and Euclidean neighborhoods. Here, we keep λ symbolic and use algebraic fibers, without requiring a norm on transfer functions.

Theorem 4: Under the irreducibility assumption on V_o, X is generically locally network identifiable in the algebraic sense from (10) if and only if

$$\dim(V_o) = \dim(V_c).$$

Proof: For a fixed $G_c \in V_c$, the fiber is

$$F^{-1}(G_c) = \{X \in V_o \mid F(X) = G_c\}.$$

Generic local network identifiability means that $\dim\left(F^{-1}(G_c)\right) = 0$ for generic $G_c \in V_c$. Since F has dense image in the Zariski closure of V_c, it follows from the fiber dimension theorem in algebraic geometry that

$$\dim(V_o) = \dim(V_c) + \dim\left(F^{-1}(G_c)\right)$$

for generic $G_c \in V_c$. Thus the generic fiber has dimension zero exactly when $\dim(V_o) = \dim(V_c)$. ∎

Remark 7: When the k free entries have no relations, excluding points where F is undefined does not change their dimension, so $\dim(V_o) = k$. To compute $\dim(V_c)$, clear the denominators in $F(X) = G_c$ and let D_F be the product of those denominators and the polynomials excluded from V_o. Form the ideal generated by the resulting polynomial equations and $1 - tD_F$. With X and t preceding the entries of G_c in a lexicographic order, the basis polynomials involving only G_c define the Zariski closure of V_c [19, Ch. 3.3]. Its dimension can be computed using [24].

Remark 8: For rational constraints on X, use the admissible subset of the constraint variety as V_o and its actual dimension. If it has several irreducible components, apply the dimension criterion to each component and the closure of its image. Repeated entries are represented by one variable.

Remark 9: If an identifiable part of the predictor determines a closed-loop transfer function, the same fiber analysis applies to that transfer function.

The map F includes the cases with fixed $Q_r = I$ or $R_y = I$ studied in [3], [15]. It also permits unknown input and measurement transfer functions and a direct term S_{yr}, beyond the fixed selection matrices of [25].

Since $K = \mathbb{C}(\lambda)$ has characteristic zero, generic local network identifiability in the algebraic sense is equivalent, in the unconstrained case, to the Jacobian of the entries of F with respect to X having rank k over $K(X)$. With constraints, the differential is restricted to directions allowed by the constraints at a generic smooth point and must have rank $\dim(V_o)$. Here, we focus on a Gröbner basis formulation.

VI. Subnetworks

Suppose the nodes of interest are w_k^A in $w_k = (w_k^A, w_k^B)$. Partition M accordingly:

$$M = \begin{bmatrix} P & Q_r & Q_e \\ -R_y & S_{yr} & S_{ye} \\ 0 & I & 0 \end{bmatrix} = \begin{bmatrix} P_A & P_{AB} & Q_{Ar} & Q_{Ae} \\ P_{BA} & P_B & Q_{Br} & Q_{Be} \\ -R_{yA} & -R_{yB} & S_{yr} & S_{ye} \\ 0 & 0 & I & 0 \end{bmatrix}.$$

Assume that the block P_A is invertible. The equations for w_k^A and y_k are

$$P_A w_k^A = \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} \begin{bmatrix} w_k^B \\ r_k \end{bmatrix} + Q_{Ae} e_k$$

$$y_k = R_{yA} w_k^A + \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix} \begin{bmatrix} w_k^B \\ r_k \end{bmatrix} + S_{ye} e_k.$$

If w_k^B is measured, (w_k^B, r_k) is the measured input to the A-subnetwork. Its input and direct-measurement matrices are $\begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix}$ and $\begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix}$, respectively. Thus its closed-loop transfer function is

$$R_{yA} P_A^{-1} \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} + \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix} = G_c.$$

Alternatively, suppose the following combinations of boundary signals are measured:

$$\begin{bmatrix} \tilde{w}_k^B \\ \hat{w}_k^B \end{bmatrix} = \begin{bmatrix} -P_{AB} \\ R_{yB} \end{bmatrix} w_k^B,$$

Then the A-subnetwork equations become

$$P_A w_k^A = \begin{bmatrix} I & 0 & Q_{Ar} \end{bmatrix} \begin{bmatrix} \tilde{w}_k^B \\ \hat{w}_k^B \\ r_k \end{bmatrix} + Q_{Ae} e_k$$

$$y_k = R_{yA} w_k^A + \begin{bmatrix} 0 & I & S_{yr} \end{bmatrix} \begin{bmatrix} \tilde{w}_k^B \\ \hat{w}_k^B \\ r_k \end{bmatrix} + S_{ye} e_k.$$

With $(\tilde{w}_k^B, \hat{w}_k^B, r_k)$ as measured input, the closed-loop transfer function is

$$R_{yA} P_A^{-1} \begin{bmatrix} I & 0 & Q_{Ar} \end{bmatrix} + \begin{bmatrix} 0 & I & S_{yr} \end{bmatrix} = G_c$$

which involves no transfer functions of the B-subnetwork.

For informativity, excitation through w_k^B must still be computed in the full network. Keep the first three block rows of M above and the measured-input row $\begin{bmatrix} 0 & 0 & I & 0 \end{bmatrix}$. For input (w_k^B, r_k), add the row $\begin{bmatrix} 0 & -I & 0 & 0 \end{bmatrix}$; for $(\tilde{w}_k^B, \hat{w}_k^B, r_k)$, add the rows $\begin{bmatrix} 0 & P_{AB} & 0 & 0 \\ 0 & -R_{yB} & 0 & 0 \end{bmatrix}$ instead. A boundary input that does not enter either subnetwork equation can be omitted

together with its predictor-input row; identically zero transformed signals can be omitted in the same way. The two choices may also be combined componentwise according to which boundary signals are measured.

VII. Examples

The following examples illustrate the rank and fiber criteria.

Example 1: Consider the network in Figure 1. The noise

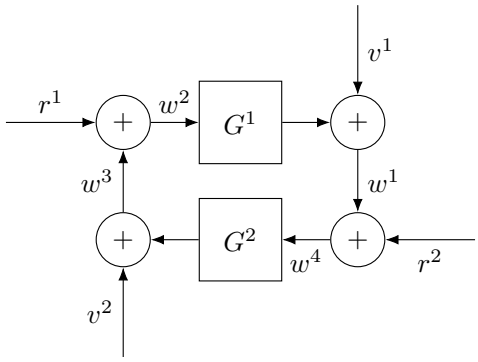


Fig. 1. Block diagram for a simple network.

inputs v_k^1, v_k^2 and filtered external inputs r_k^1, r_k^2 satisfy $v_k^1 = H^1 e_k^1$, $v_k^2 = H^2 e_k^2$ and $r_k^1 = Q^1 r_k$, $r_k^2 = Q^2 r_k$. Assume that $H^1 \not\equiv 0$ and $H^2 \not\equiv 0$. The single input r_k enters the loop at two nodes. We are interested in estimating (G^1, Q^1), and use a predictor for predicting w_k^1. We can measure (w_k^3, r_k) in addition to w_k^1, and therefore the predictor input is $z_k = (w_k^1, w_k^3, r_k)$. We obtain the following equations

$$\begin{bmatrix} w_k^1 \\ w_k^2 \\ w_k^3 \\ w_k^4 \end{bmatrix} = \begin{bmatrix} 0 & G^1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & G^2 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} w_k^1 \\ w_k^2 \\ w_k^3 \\ w_k^4 \end{bmatrix} + \begin{bmatrix} 0 & H^1 & 0 \\ Q^1 & 0 & 0 \\ 0 & 0 & H^2 \\ Q^2 & 0 & 0 \end{bmatrix} \begin{bmatrix} r_k \\ e_k^1 \\ e_k^2 \end{bmatrix}$$
$$z_k = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} w_k^1 \\ w_k^2 \\ w_k^3 \\ w_k^4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r_k. \tag{11}$$

Here $\det P = 1 - G^1 G^2$, so P is generically invertible. There are three disjoint paths: $(\mathcal{U}_2, \mathcal{W}_1, \mathcal{Z}_1)$, $(\mathcal{U}_3, \mathcal{W}_3, \mathcal{Z}_2)$, and $(\mathcal{U}_1, \mathcal{Z}_3)$. Hence we have informativity under the standing well-posedness and stability assumptions, assuming that the spectrum for (r_k, e_k^1, e_k^2) is positive definite for almost all frequencies. We can also verify this by directly looking at the transfer function

$$\Pi = \begin{bmatrix} \frac{G^1(Q^1+G^2Q^2)}{1-G^1G^2} & \frac{H^1}{1-G^1G^2} & \frac{G^1H^2}{1-G^1G^2} \\ \frac{G^2(G^1Q^1+Q^2)}{1-G^1G^2} & \frac{G^2H^1}{1-G^1G^2} & \frac{H^2}{1-G^1G^2} \\ 1 & 0 & 0 \end{bmatrix}$$

which has determinant $H^1H^2/(1 - G^1G^2)$. Hence, both the rank and the generic rank are full under well-posedness and the nonzero noise-filter assumptions.

The closed loop system transfer function from (w_k^3, r_k) to w_k^1 is given by

$$\begin{bmatrix} G_c^1 & G_c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -G^1 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 \\ 1 & Q^1 \end{bmatrix} = \begin{bmatrix} G^1 & G^1Q^1 \end{bmatrix}.$$

The dimensions of the domain and the image closure are $\dim V_c = \dim V_o = 2$ and we conclude that

$$(G^1, Q^1) = (G_c^1, G_c^2/G_c^1), \qquad G_c^1 \not\equiv 0.$$

Here, nonzero means not identically zero as a rational function. Hence, we have generic network identifiability, and therefore also generic local network identifiability. Uniqueness fails at $G_c = (0, 0)$, where $G^1 = 0$ and Q^1 is arbitrary. This zero value is permitted by the structural pattern but is nongeneric. Thus, identifiability does not hold for every admissible closed loop transfer function.

Example 2: Let us now consider the case where we are only interested in estimating G^1 and Q^1 is known. Using the same measured signals, we can form $\xi_k = w_k^3 + Q^1(q) r_k$. The network equation becomes

$$w_k^1 = G^1(q)\xi_k + H^1(q)e_k^1.$$

Under the assumed innovation representation, let $V(q)$ be the predictor transfer function from ξ_k to $\hat{w}_k^1$. The predictor is

$$\hat{w}_k^1 = W_y(q) w_k^1 + V(q)\xi_k, \qquad G^1 = (1 - W_y)^{-1} V.$$

Equivalently, the original predictor columns are (W_y, V, VQ^1). Following Remark 1, it suffices to establish informativity of (w_k^1, ξ_k), rather than positive definiteness of the spectrum of all three original predictor inputs.

With $e^2 = 0$, the transfer function matrix from (r, e^1) to (w^1, ξ) has determinant

$$-\frac{H^1(Q^1 + G^2Q^2)}{1 - G^1G^2}.$$

With $r = 0$, the corresponding determinant for inputs (e^1, e^2) is

$$\frac{H^1H^2}{1 - G^1G^2}.$$

In either case, a positive-definite spectrum of the active inputs for almost all frequencies ensures informativity when the displayed determinant is not identically zero, under the standing well-posedness and stability assumptions. Hence, known Q^1 allows reduced excitation by combining predictor inputs, rather than simply discarding an unknown predictor column. This illustrates how prior knowledge can reduce the excitation needed to identify G^1.

Example 3: Consider three unmeasured node signals $w_k = (w_k^1, w_k^2, w_k^3)$, measured inputs $r_k = (r_k^1, r_k^2, r_k^3)$, and the measured output y_k. Let G, A, and H be unknown open-loop transfer functions, and take

$$P = \begin{bmatrix} 1 & -G & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q_r = \begin{bmatrix} 1 & 0 & 0 \\ 0 & A & 1 \\ 0 & A & 0 \end{bmatrix}, \quad Q_e = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$
$$R_y = \begin{bmatrix} H & 0 & A \end{bmatrix}, \qquad S_{yr} = 0, \qquad S_{ye} = 0.$$

Thus $w_k^2 = Ar_k^2 + r_k^3$, $w_k^1 = Gw_k^2 + r_k^1 + e_k$, $w_k^3 = Ar_k^2$, and $y_k = Hw_k^1 + Aw_k^3$. The same unknown A occurs in two entries of Q_r and one entry of R_y. The closed-loop transfer function from r_k to y_k is

$$\begin{aligned} G_c = R_y P^{-1} Q_r &= \begin{bmatrix} H & HGA + A^2 & HG \end{bmatrix} \\ &= \begin{bmatrix} G_c^1 & G_c^2 & G_c^3 \end{bmatrix}. \end{aligned}$$

A generic fiber therefore satisfies $H = G_c^1$, $G = G_c^3/G_c^1$, and

$$A^2 + G_c^3 A - G_c^2 = 0.$$

For $G_c^1 \not\equiv 0$ and $2A + G_c^3 \not\equiv 0$, this has exactly two distinct solutions, A and $A' = -G_c^3 - A = -HG - A$. Both are rational transfer functions and, when G, H, and A are proper and stable, so is A'. The Jacobian of (G_c^1, G_c^2, G_c^3) with respect to (G, A, H) has determinant $-H(HG+2A)$, confirming generic local identifiability in the algebraic sense. The quadratic shows that the generic fiber has two points, so the network is not generically network identifiable. This fiber calculation reveals the twofold ambiguity, which the generic Jacobian-rank test alone does not distinguish from uniqueness.

For the predictor input $z_k = (y_k, r_k)$ and external signals $u_k = (r_k, e_k)$, the determinant of the transfer function matrix Π is $-H$. Thus full generic row rank, and hence generic informativity under the standing spectral and predictor assumptions, holds when $H \not\equiv 0$.

The Gröbner basis rank test gives the same result. Here Π is a polynomial 4×4 matrix and $\det P = 1$, so $D = 1$. For $k = 4$ its ideal is $\mathcal{I}_4 = \langle \det \Pi, 1 - t \rangle = \langle H, 1 - t \rangle$. With t first in the lexicographic order, the reduced basis is $\{t - 1, H\}$. Its polynomial independent of t shows that rank falls below four exactly when $H = 0$.

For the identifiability test, regard G_c^1, G_c^2, G_c^3 as output variables over $K = \mathbb{C}(\lambda)$. The repeated unknown A is a single variable. The ideal defining the image closure is obtained by eliminating t, G, A, H from

$$\langle G_c^1 - H,\ G_c^2 - HGA - A^2,\ G_c^3 - HG,\ 1 - t \rangle.$$

For example, use lexicographic order $t > G > A > H > G_c^3 > G_c^2 > G_c^1$. The resulting elimination ideal in $K[G_c^1, G_c^2, G_c^3]$ is zero: when $G_c^1 \neq 0$, setting $H = G_c^1$ and $G = G_c^3/G_c^1$ leaves the quadratic equation for A above, which has a root over the algebraic closure of K. Hence $\dim(V_c) = 3 = \dim(V_o)$, as required by the identifiability criterion. The two roots give the generic twofold ambiguity.

VIII. Computational Examples

The following calculations use the Gröbner basis tests on a five-node network. The MATLAB scripts and reproduction instructions are in the accompanying supplementary archive `supplementary_matlab_code.zip`. Its `README.md` identifies the two example scripts and three helper files. For independent free entries, the path test avoids symbolic elimination. A generic Jacobian-rank test decides whether fibers are finite; elimination can additionally reveal the number of solutions and exceptional sets, as in Example 3. Gröbner calculations can be costly as the number of unknowns grows; the timings below describe these instances, not a scaling law.

Example 4: Consider the dynamical network defined by (P, Q_r, R_y, S_{yr}), where $P = I - G$, $S_{yr} = 0$ and

$$G = \begin{bmatrix} 0 & G^{12} & 0 & 0 & 0 \\ 0 & 0 & G^{23} & 0 & 0 \\ 0 & 0 & 0 & G^{34} & 0 \\ 0 & 0 & 0 & 0 & G^{45} \\ G^{51} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_r = \begin{bmatrix} Q^1 & 1 & 0 & 0 & 0 \\ 0 & Q^2 & 1 & 0 & 0 \\ 0 & 0 & Q^3 & 1 & 0 \\ 0 & 0 & 0 & Q^4 & 1 \\ Q^5 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_y = \begin{bmatrix} R^1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Here $\det P = 1 - G^{12}G^{23}G^{34}G^{45}G^{51}$, so P is generically invertible. The eleven unknown open-loop transfer functions give $\dim V_o = 11$. Elimination gives $\dim V_c = 11$, and hence generic local network identifiability in the algebraic sense. With $Q_e = I$ and $S_{ye} = 0$, the generic row rank of Π is eight, the number of predictor inputs; rank falls when $R^1 = 0$. Theorem 1 therefore gives generic informativity if the spectrum of (r_k, e_k) is positive definite for almost all frequencies and the predictor is stable. The image-dimension computation took 631 seconds in one run with MATLAB R2025a and Symbolic Math Toolbox 25.1 on an AMD Ryzen 7 250 with 31 GB RAM.

Example 5: In the same network, consider the subnetwork with unknown transfer functions (G^{12}, Q^1, Q^2, R^1). Measure $y_k = (R^1(q)w_k^1, w_k^2)$ and the boundary signal $\tilde{w}_k^B = G^{23}w_k^3$. Its closed-loop transfer function is

$$F(X) = \begin{bmatrix} R^1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -G^{12} \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 & Q^1 & 1 & 0 \\ 1 & 0 & Q^2 & 1 \end{bmatrix}$$

for input $(\tilde{w}_k^B, r_k^1, r_k^2, r_k^3)$. The displayed P_A has determinant one. With $Q_e = I$ and $S_{ye} = 0$, the generic row ranks of Π are eight, six, and five when the active measured inputs are, respectively, $(r_k^1, r_k^2, r_k^3, r_k^4, r_k^5)$, (r_k^1, r_k^2, r_k^3), and (r_k^1, r_k^2). In each case, a stable predictor and a positive-definite spectrum of the active external signals for almost all frequencies give generic informativity. The image dimension is four both for the displayed $F(X)$ and after its last column is removed, so generic local network identifiability also holds when $r_k^3 = 0$. The first and last columns of $F(X)$ coincide.

IX. Conclusions

Positive definiteness of the external-signal spectrum and full generic row rank of the transfer function to the predictor inputs give a sufficient condition for generic informativity. Vertex-disjoint paths determine that rank for independent free entries; Gröbner bases also handle known or constrained entries. The fiber-dimension test gives generic finite ambiguity for the broader network model with unknown input and measurement transfer functions. Example 3 has two admissible networks in a generic fiber, which the Jacobian-rank test alone does not count. The algebraic genericity statements do not certify a specified realization, and symbolic elimination can be expensive even for small networks.

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